

Let $\sup_{n \in \mathbb{N}} \{ \deg(p_n) : n \in \mathbb{N} \} \leq M$, $M > 0$ & $M \in \mathbb{N}$.

The $\deg(p_n) \leq M \Rightarrow \{p_n\} \subseteq \mathcal{P}_M :=$ Vector space
of pol^n , $\deg \leq M$.

$$\mathcal{P}_M \cong \mathbb{R}^{M+1}$$

$$p_n \xrightarrow{\|\cdot\|_\infty} f \in \mathcal{P}_M \Rightarrow f \text{ is a } \text{pol}^n. \quad \textcircled{\Rightarrow \Leftarrow}$$

Hence $\nexists M > 0$.